

Version-Aware Normalization of Findings Against MITRE ATT&CK v19

Typed Data Models and Revocation-Resolving Technique Lookup for Security Tooling

github.com/poojakira/attack-v19-core - public, default branch main.

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

165

Tests passed

63.83%

Statement coverage

113

Test functions

11

Test files

SYSTEM VIEW

Architecture and Threat-Control Flow



Research Contribution

A typed ATT&CK v19 data layer that normalizes findings across version churn by resolving revoked/renamed technique IDs through a maintained remap table over SHA-256-verified STIX data, with indexed lookup for downstream security tooling.

Verified Evidence

EV-01	Revoked technique IDs: 22	EV-02	Total remaps: 29
EV-03	New techniques in documented v19 migration: 48	EV-04	165 tests passed.
EV-05	63.83% statement coverage; CI gate is 60%.	EV-06	113 test functions across 11 test files; parametrization expands these to 165 executed tests.
EV-07	Current repository mapping evidence retains:	EV-08	22 revoked technique IDs
EV-09	29 total remaps	EV-10	48 new techniques for the documented v19 migration surface

Method and Control Path

01	Download/validate ATT&CK STIX data with SHA-256 verification.
02	Parse techniques/tactics/platform metadata into typed structures.
03	Resolve deprecated/revoked IDs through the v19 remap table.
04	Build in-memory indexes for ID, tactic, platform, and keyword lookup.
05	Regression-test migration and data-loader behavior across supported Python versions.

Limitations and Residual Risk

- No additional limitation text was found in the research brief. Consult the README before generalizing results.

Reproducibility

Expected current CI evidence: 165 passed, 63.83% coverage.

Evidence Provenance

Claim discipline Metrics come from the repository's verified metrics and research brief.
Scope discipline Benchmarks are reported only at their documented fixture, dataset, or test scope.
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